General Bite #7: The HF Transceiver Block Diagram

How an SSB transceiver is put together: balanced modulator, sideband filter, mixer, product detector, DSP, SDR I/Q.

Group G7C · 14 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

The SSB Generation Chain

To make an SSB signal, the transmitter follows a fixed recipe.

Stage	What it does	Q
Balanced modulator	**Produces double-sideband modulated RF** (G7C02). Carrier is suppressed; both sidebands present.	G7C02
Filter (after balanced mod)	**Selects one of the sidebands** (G7C01). USB or LSB drops out, the other goes on.	G7C01
Impedance matching transformer (output)	**Presents the desired impedance to the transmitter and feed line** (G7C03).	G7C03

The SSB Reception Chain

Product detector = **Used in a single-sideband receiver to extract the modulated signal** (G7C04).
A product detector is just a mixer fed with the IF signal and a beat-frequency oscillator (BFO) set to reinsert the carrier at the IF stage. The audio appears at the output.

Filter Vocabulary

Term	Definition	Q
Insertion loss	**Filter's attenuation inside its passband** (G7C07). Lower = better.	G7C07
Ultimate rejection	**Filter's maximum ability to reject signals outside its passband** (G7C13).	G7C13
Cutoff frequency	**The frequency above which a low-pass filter's output power is less than half the input power** (G7C12). The −3 dB point.	G7C12
Band-pass filter bandwidth	Measured between **upper and lower half-power** frequencies (G7C14).	G7C14

DDS, DSP, and SDR

Question	Answer	Q
Direct digital synthesizer (DDS) characteristic	**Variable output frequency with the stability of a crystal oscillator** (G7C05).	G7C05
DSP filter advantage	**A wide range of filter bandwidths and shapes can be created** (G7C06). Software, not hardware, defines the response.	G7C06
Phase between I and Q in SDR	**90 degrees** (G7C09). Hence the names "in-phase" (I) and "quadrature" (Q).	G7C09
Advantage of I/Q modulation in SDR	**All types of modulation can be created with appropriate processing** (G7C10). One hardware chain, every mode.	G7C10
Functions an SDR performs in software	Filtering + detection + modulation – **all of the above** (G7C11).	G7C11

Receiver Sensitivity Factors

What affects receiver sensitivity? Input-amplifier gain + demodulator-stage bandwidth + input-amplifier noise figure – **all of the above** (G7C08). The whole front-end chain matters.

Mental Cheat Sheet

- Balanced mod output = DSB.
- SSB filter selects one sideband.
- Impedance-matching transformer presents target Z to TX.
- Product detector recovers SSB audio.
- Insertion loss = passband attenuation.
- Ultimate rejection = stopband attenuation depth.
- LP filter cutoff = where output power drops to half.
- BP filter bandwidth = upper to lower half-power.
- DDS = variable freq with crystal stability.
- DSP filter = arbitrary response in software.
- SDR I/Q = 90° apart.
- I/Q modulation = any mode in software.
- SDR does filtering, detection, modulation.
- RX sensitivity depends on gain, bandwidth, NF.

Common Traps

- Balanced modulator outputs **DSB**, not SSB (filter does the SSB).
- Product detector is for **SSB**, not test-gear spurious detection.
- Insertion loss is **inside** the passband; ultimate rejection is **outside**.
- Cutoff = **half-power** point, not where the curve goes flat.
- DDS stability = crystal-quality, **not** "narrow tuning."
- I and Q are **90°** apart, not 45° or 180°.
- SDR can synthesize **all** modulation types – the trap answers say "minimum detectable signal reduced."
- Sensitivity depends on **the whole chain**, not just one parameter.

Takeaway: SSB out: carrier suppressed at balanced mod, one sideband selected by filter. SSB in: product detector with a BFO. Filter specs: insertion loss in passband, ultimate rejection in stopband, cutoff at half-power. DDS gives variable frequency at crystal stability. SDR uses 90°-offset I/Q signals processed in software to produce any modulation type.

Bite 7 Question Bank (G7C)

G7C01. What circuit is used to select one of the sidebands from a balanced modulator?
A. Carrier oscillator.
B. Filter.
C. IF amplifier.
D. RF amplifier.

G7C02. What output is produced by a balanced modulator?
A. Frequency modulated RF.
B. Audio with equalized frequency response.
C. Audio extracted from the modulation signal.
D. Double-sideband modulated RF.

G7C03. What is one reason to use an impedance matching transformer at a transmitter output?
A. To minimize transmitter power output.
B. To present the desired impedance to the transmitter and feed line.
C. To reduce power supply ripple.
D. To minimize radiation resistance.

G7C04. How is a product detector used?
A. Used in test gear to detect spurious mixing products.
B. Used in a transmitter to perform frequency multiplication.
C. Used in an FM receiver to filter out unwanted sidebands.
D. Used in a single sideband receiver to extract the modulated signal.

G7C05. Which of the following is characteristic of a direct digital synthesizer (DDS)?
A. Extremely narrow tuning range.
B. Relatively high-power output.
C. Pure sine wave output.
D. Variable output frequency with the stability of a crystal oscillator.

G7C06. Which of the following is an advantage of a digital signal processing (DSP) filter compared to an analog filter?
A. A wide range of filter bandwidths and shapes can be created.
B. Fewer digital components are required.
C. Mixing products are greatly reduced.
D. The DSP filter is much more effective at VHF frequencies.

G7C07. What term specifies a filter's attenuation inside its passband?
A. Insertion loss.
B. Return loss.
C. Q.
D. Ultimate rejection.

G7C08. Which parameter affects receiver sensitivity?
A. Input amplifier gain.
B. Demodulator stage bandwidth.
C. Input amplifier noise figure.
D. All these choices are correct.

G7C09. What is the phase difference between the I and Q RF signals that software-defined radio (SDR) equipment uses for modulation and demodulation?
A. Zero.
B. 90 degrees.
C. 180 degrees.
D. 45 degrees.

G7C10. What is an advantage of using I-Q modulation with software-defined radios (SDRs)?
A. The need for high resolution analog-to-digital converters is eliminated.
B. All types of modulation can be created with appropriate processing.
C. Minimum detectible signal level is reduced.
D. Automatic conversion of the signal from digital to analog.

G7C11. Which of these functions is performed by software in a software-defined radio (SDR)?
A. Filtering.
B. Detection.
C. Modulation.
D. All these choices are correct.

G7C12. What is the frequency above which a low-pass filter's output power is less than half the input power?
A. Notch frequency.
B. Neper frequency.
C. Cutoff frequency.
D. Rolloff frequency.

G7C13. What term specifies a filter's maximum ability to reject signals outside its passband?
A. Notch depth.
B. Rolloff.
C. Insertion loss.
D. Ultimate rejection.

G7C14. The bandwidth of a band-pass filter is measured between what two frequencies?
A. Upper and lower half-power.
B. Cutoff and rolloff.
C. Pole and zero.
D. Image and harmonic.

Answer Key

G7C01: B	G7C02: D	G7C03: B	G7C04: D
G7C05: D	G7C06: A	G7C07: A	G7C08: D
G7C09: B	G7C10: B	G7C11: D	G7C12: C
G7C13: D	G7C14: A		